

net. In contrast, modern American conservatism, associated with the Republican Party, aligns itself with individual liberty, limited government, free-market capitalism, and traditional social values. These differing core beliefs create distinct partisan biases, shaping how individuals interpret facts, prioritize issues, and frame political debates.

2.2 Bias Mitigation Strategies in LLMs

Large language models (LLMs) have demonstrated remarkable abilities in language generation, but they also exhibit significant biases that can negatively impact society (Gupta et al., 2022; Sheng et al., 2019; Huang et al., 2023). (Navigli et al., 2023) define social biases in LLMs as prejudices, stereotypes, and discriminatory attitudes directed against particular groups of people. These biases manifest across dimensions such as gender, race, social class, disability, nationality, and religion. Prior studies have shown that such biases often originate from training corpora and the word embeddings derived from pre-trained models (Sun et al., 2019).¹

To counteract these risks, a wide range of strategies have been proposed to mitigate bias in LLMs. One class of approaches centers on prompt-based interventions. For example, Gallegos et al. (2024) introduced a self-debiasing technique in which models are prompted to identify potential biases before answering a question in a zero-shot setting. Similarly, few-shot prompting and chain-of-thought reasoning have been used to guide models toward less biased responses (Dwivedi et al., 2023; Huang et al., 2023). While promising, such approaches often struggle to generalize across bias types and require substantial human supervision. Another class of methods emphasizes data augmentation. Zhao et al. (2018) proposed gender-swapped datasets to mitigate bias in embeddings, while Zmigrod et al. (2019) employed counterfactual augmentation by reversing gendered pronouns in Wikipedia. Park et al. (2018) explored transfer learning, leveraging unbiased datasets during fine-tuning to reduce bias. In contrast to these strategies, other work generates unbiased data by prompting LLMs to create biased story-judgment pairs, then revising those judgments in a gender-neutral manner. This two-step process produces unbiased training data that can be used for retraining and more effective debiasing.

¹Which is essentially training data, at one additional layer of remove.

Besides data and prompt-centric approaches, fine-tuning alignment methods have also emerged as bias mitigation strategies. Reinforcement Learning from Human Feedback (RLHF) has been a widely adopted paradigm, where a reward model trained on human evaluations guides a policy model via Proximal Policy Optimization (PPO) (Ouyang et al., 2022). RLHF has shown effectiveness in aligning model outputs with human preferences and reducing bias, but it faces challenges including reward hacking, instability, mode collapse, and the need for a separate reward model (Casper et al., 2023). As an alternative, Direct Preference Optimization (DPO) has recently been introduced (Rafailov et al., 2023). Instead of relying on a reward model, DPO directly optimizes the policy model by maximizing the log-likelihood of preferred responses and minimizing that of dispreferred ones under given prompts. This closed-form optimization avoids reinforcement learning altogether, making training more stable and efficient while mitigating risks such as reward hacking. DPO performs as well as or better than PPO in tasks like sentiment-controlled text generation, demonstrating its potential as a robust bias mitigation technique. However, the extent to which various alignment methods penetrate the model’s internal states and processes is still an open question, as bias that appears superficially to be ameliorated can reappear under the right conditions (Hofmann et al., 2024; Zimmerman et al., 2025).

2.3 Automatic Data Generation

Manually constructing alignment datasets is not only time-consuming and labor-intensive, but it can also risk introducing toxic content (Zhao et al., 2018). To address these challenges, recent work has explored prompting LLMs to generate synthetic datasets, typically beginning with a small set of human-annotated seed examples and then expanding them through few-shot prompting (Sun et al., 2019; Xu et al., 2023). However, such methods often suffer from limited diversity, as the generated data tend to closely mirror the initial seed examples (Lucy and Bamman, 2021). Another line of research generates alignment data by transforming existing datasets (Wang et al., 2024; Sanh et al., 2021). In contrast, our novel data generation framework does not rely on a handful of seed examples or common transformations of pre-existing datasets (e.g., swapping nouns or reversing gendered pronouns). Instead, it synthesizes a wide and politi-

cally inclusive range of narratives (e.g. retaining discussion of Liberals and Conservatives), reducing dependence on potentially biased or limited source material.

3 RQ1: Measuring us-versus-them biases in LLMs

3.1 Method

To investigate the extent of us versus them biases across five LLMs from different families (e.g., GPT-4.1, Deepseek-3.1, Gemma-2.0, Grok-3.0, and Llama-3.1). To assess the social identity biases for each language model, we generated a total of two thousand sentences for ingroup sentence (“We are”) and outgroup sentence (“They are”) prompting with in their default settings, which are associated with the ‘us versus them’ dynamics (Hu et al., 2023), excluding sentences that did not pass minimal quality and diversity checks. We call sentences starting with “We are” ingroup sentences and those starting with “They are” outgroup sentences. For our experimenting models, it suffices to use the prompt “We are” or “They are” and let the model complete the sentence by repeatedly generating the next tokens. The prompt shows in Appendix A.2 in Table 5. We refer to this prompt setting as the Default Prompt (Supporting Information for A.1 examples from default prompt).

‘Us versus them’ bias score. Social Identity Theory (SIT) (Tajfel et al., 1971; Tajfel and Turner, 1979) serves as the theoretical framework for quantifying ‘us versus them’ bias score in this study. SIT explains how individuals define themselves through group memberships and how this process leads to ingroup favoritism and outgroup differentiation. The theory’s foundation rests on findings from minimal group paradigm experiments (Tajfel et al., 1971), which established that even arbitrary group distinctions are sufficient to elicit preferential treatment toward one’s own group. According to SIT, individuals categorize themselves and others into groups, identify with valued groups to derive self-esteem, and seek positive distinctiveness by engaging in social comparison that favors their ingroup over outgroups. To capture the degree of ingroup favoritism or outgroup derogation in LLM-generated sentences, scores were assigned on a 0–100 scale, reflecting the motivational principles of SIT (the specific scoring prompt is detailed

in Appendix A.2, Table 8).

For evaluation, we employed GPT-4.1 as the judging model due to its close alignment with human judgment, achieving a mean absolute error (MAE) of 12.6. A detailed reliability study comparing the Us-vs-Them scoring of various LLMs (Deepseek-3.1, GPT-4.1, Gemma-2.0, Grok-3.0, and Llama-3.1) against human scores is presented in Appendix A.3, with the specific reliability performance summarized in Table 11. Examples of model-generated sentences and their assigned bias scores are provided in Table 12. Appendix A.4, Figure 17 shows the distribution of bias scores for each LLM-generated ingroup and outgroup sentences.

To estimate ingroup solidarity (μ_{ingroup}) and outgroup hostility (μ_{outgroup}) of a model, we compute the mean us versus them bias score for each category of LLM-generated sentences. Formally:

$$\mu_{\text{ingroup}} = \frac{1}{N_{\text{in}}} \sum_{i=1}^{N_{\text{in}}} s_i^{(\text{in})}, \mu_{\text{outgroup}} = \frac{1}{N_{\text{out}}} \sum_{j=1}^{N_{\text{out}}} s_j^{(\text{out})} \quad (1)$$

where $s_i^{(\text{in})}$ is the us versus them score for the i -th ingroup sentence (starting with “We are”) and $s_j^{(\text{out})}$ is the score for the j -th outgroup sentence (starting with “They are”), and N_{in} and N_{out} are the total number of ingroup and outgroup sentences.

Allotaxonomy. To compare word usage between ingroup (“We are”) and outgroup (“They are”) sentences, we employed the allotaxonomic rank-rank histogram (Dodds et al., 2023) and rank-turbulence divergence as analytical tools. We first extracted 1-grams (continuous sequences of non-whitespace characters) from both sentence groups. For each group, we constructed a frequency-ranked list of these 1-grams. We then used the allotaxonometer to merge the word types from both corpora into a unified lexicon and compute logarithmic rank-rank pairs ($\log_{10} r_1, \log_{10} r_2$), where r_1 and r_2 are the ranks of a word in the ingroup and outgroup lists, respectively. These pairs were binned and plotted in uniformly spaced logarithmic axes. Bins to the left or right of the vertical axis represented words more frequently used in the respective group. To quantify divergence between the two groups, we applied the rank-turbulence divergence metric using a tunable parameter $\alpha = 1/3$, as detailed in (Dodds et al., 2023).

Sentiment shifts. We also measured the sentiment dynamics of ingroup and outgroup sentences using dictionary-based sentiment analysis. This

method is inherently sensitive to the dictionary employed, as sentiment lexicons are often static and constructed once for general use. Such static design can be problematic when the meaning of words shifts over time or when words adopt different connotations in specific contexts (Loughran and McDonald, 2011). Word shift graphs (Gallagher et al., 2020a) provide a transparent diagnostic tool to reveal such measurement issues. To visualize the sentiment shifts between ingroup (“we are”) and outgroup (“they are”) sentences, we applied the labMT sentiment dictionary (Dodds et al., 2011). In labMT, each word is assigned a happiness score on a 1–9 scale, ranging from sad to happy, with neutral words averaging around 5. For word shift graphs in Figure 3, a reference value of 5 was used, and words with sentiment scores between 4 and 6 were removed (filtering out neutral words). The resulting graphs also include cumulative contribution and text size diagnostic plots in the bottom left and right corners, respectively. We measured the difference in average sentiment (Φ_{avg}) between ingroup and outgroup sentences, ranking words by their absolute contribution to this difference.

3.2 Results and Discussion

Table 1 presents the ingroup solidarity ($\mu_{ingroup}$) and outgroup hostility ($\mu_{outgroup}$) scores under the default prompt setting for each LLM. Among the five recent models (DeepSeek-3.1, GPT-4.1, Gemma-2.0, Grok-3.0, and LLaMA-3.1), all exhibit ingroup solidarity, while four (except DeepSeek-3.1) show outgroup hostility. This aligns with prior findings on LLMs’ social identity bias (Hu et al., 2023). Ingroup favoritism generally exceeds outgroup hostility, reflecting human-like tendencies to favor the ingroup more than derogating the outgroup. GPT-4.1 shows the strongest ingroup bias (84.83) with low outgroup hostility (45.21) and low variability, indicating consistent favoritism. DeepSeek-3.1 and Grok-3.0 show moderate ingroup bias (74.79 and 69.59) with slightly lower hostility, while Gemma-2.0 shows near balance but high variability. LLaMA-3.1 exhibits the lowest ingroup bias (53.06) and highest outgroup variability (Std = 24.19), indicating unstable outgroup evaluations.

To examine semantic differences between ingroup (“We are”) and outgroup (“They are”) sentences in LLM outputs, we performed hierarchical clustering on sentence embeddings using cosine distance and the ‘average’ linkage method.

Table 1: Ingroup solidarity ($\mu_{ingroup}$) and outgroup hostility ($\mu_{outgroup}$) of each LLM (Deepseek-3.1, GPT-4.1, Gemma-2.0, Grok-3.0, and Llama-3.1) under default prompt settings. Std denotes the standard deviation of the us versus them scores.

Model	Bias Type	Score	Std
DeepSeek-3.1	$\mu_{outgroup}$	60.53	11.31
	$\mu_{ingroup}$	74.79	12.25
GPT-4.1	$\mu_{outgroup}$	45.21	10.71
	$\mu_{ingroup}$	84.83	6.68
Gemma-2.0	$\mu_{outgroup}$	63.25	19.68
	$\mu_{ingroup}$	60.20	10.75
Grok-3.0	$\mu_{outgroup}$	60.43	12.19
	$\mu_{ingroup}$	69.59	13.79
Llama-3.1	$\mu_{outgroup}$	63.59	24.19
	$\mu_{ingroup}$	53.06	10.42

As shown in Appendix A.5, Figure 19, ingroup and outgroup sentences form distinct clusters, indicating clear semantic separation driven by the ingroup/outgroup cue.

The allotaxonomic rank-rank histogram in Figure 2, reveals clear differences in word usage between ingroup and outgroup sentences. Near the bottom of the plot, discrete lines represented exclusive words—those appearing only in one group. Two additional lines just above these marked words that occurred only once or twice in the other group. Words near the top of the histogram, such as “are,” were highly frequent in both groups ($r_{RT,1} = r_{RT,2} = 1$). Words like “defenders” and “champions” ranked 17th and 14th in the ingroup list but reversed in the outgroup list, illustrating semantic role shifts. We observed turbulence in word rankings beginning around rank $r = 1000$, where more distinguishing words began to appear. Words farther from the central axis showed greater rank divergence, indicating their strong association with only one group. The right side of Figure 2 displays the rank-turbulence divergence graph. This plot ranks the top 30 words by decreasing divergence contribution ($\delta D_{1/3,\tau}^R$), with orientation reflecting the group in which each word held a higher rank. Words such as “we” and “they” are the largest contributors to divergence, followed by “our” and “their.” Although rare exclusive words like “equality” and “blame” also reflected strong group identity, their low frequency reduced their impact on the divergence measure. Figure 2 shows that LLMs tend to generate more positive, self-affirming lan-